

Clara Justino Limaco

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ABOUT ME

Software Engineer with 9 years of experience and a Master's Degree in Internet of Things, currently pursuing a Master's in Artificial Intelligence with a focus on machine learning and multi-agent systems. I have worked across diverse industries, including education technology, e-commerce, mining, the public sector (Public Prosecutor's Office and tax administration), and broadcasting. My work spans backend and full-stack development, DevOps, and cloud (AWS) and on premises infrastructure, with a background primarily focused on Python, JavaScript/TypeScript, and Java, with most of my experience centered around Django, Node.js, and React.js, while also having had the opportunity to work with other frameworks and technologies. I am passionate about combining software engineering with AI to build scalable, data-driven solutions.

Location: Barcelona, Spain
Valid legal residence: Eligible for Highly Qualified Professional authorization (Spanish Law 14/2013)

WORK EXPERIENCE

BACKEND DEVELOPER – AVENUECODE - NEWSELA CLIENT – 06/01/2024 – Current – REMOTE

- Started as a backend Python developer on the CMS metadata team, working mainly on Langston, Newsela's academic standards microservice (Django, Celery, Prefect, PostgreSQL, Neo4j). I owned its pipeline that ingests third-party (Certica) standards updates into the graph database through event-driven processing, automatic retries and scheduled runs. I also built the admin workflows for standards replacements, crosswalks and bulk tagging, with apply and revert actions run as Celery tasks.
- Built metadata and tagging features across Langston and Alexandria, Newsela's editorial CMS: extracting standards from lesson text and tagging them on lessons, a Label Studio integration for reviewing recommended concepts and topics, and an Alexandria API option to skip publishing to Langston. I also removed the old standards schema from the Django monolith and set up feature test environments with Terraform.
- Took on frontend work as my scope grew, first in Alexandria and the metadata manager (React/TypeScript), then in Core, Newsela's student and teacher platform. In Core and its design-system component library, I built course and lesson pages and shared components, and fixed accessibility, mobile layout and security vulnerability issues.
- I ended contributing to Whitman, the new CMS platform that moves Alexandria onto a content editor shared with Core, with real-time collaboration (Fastify, Drizzle, Yjs)
- Adopted Claude Code to improve my development workflow: running parallel agent sessions in git worktrees with hooks that lint every AI edit, building a custom triage skill for on-call investigations through the Datadog and Jira MCP servers, and applying spec-driven development on Whitman.

Used Technologies: Python, Django, Celery, Prefect, Javascript, TypeScript, React, Fastify, NodeJs, PostgreSQL, Neo4j, GraphQL, AWS, Terraform, Datadog

SOFTWARE DEVELOPER – STRACON TECH – 31/10/2022 – 31/12/2023 – PERU

StockControl: Mining warehouse and supplies management
- Developed backend web services to track inventory of critical mining supplies, spare parts, and materials across warehouses, supporting continuity of operations at site.
- Deployed and maintained the system on on-premises Windows Server infrastructure.

Pfinos: Mineral fines production forecasting
- Built backend web services for a platform that estimates and tracks fine metal content (“finos”) from mining production, supporting operational and planning decisions.
- Integrated machine learning prediction models into the backend, exposing production forecasts through APIs for engineering and management teams.
- Managed deployment on Windows Server.

Proximity: UWB-based safety system for mining operations
- Developed the backend for a real-time proximity detection system using UWB (Ultra-Wideband) technology to monitor distances between heavy equipment and personnel, helping prevent collisions and improve on-site safety.

Used Technologies: Nodered, Redis, Django, PostgreSQL, Docker, Python, Javascript

PYTHON DEVELOPER – DEEPPIT TECHNOLOGY – 28/02/2022 – 31/10/2022 – LIMA, PERU

Pfinos: Blast planning and fragmentation optimization platform
- Led requirements gathering with mining engineers and stakeholders to define the platform’s scope, translating mine planning and blasting workflows into technical specifications.

- Designed and developed REST APIs to support the mine planning and blasting processes, including management of blast designs and operational data.
- Integrated a machine learning solution into the platform, enabling data-driven predictions to support blasting decisions.

Technologies: Python, Django, PostgreSQL

DEVOPS ANALYST – MINISTERIO PUBLICO - FISCALIA DE LA NACION – 31/08/2020 – 31/10/2022 – PERU

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1. Implementation of continuous integration with Gitlab-CI tool:
    - Development of pipelines for backend and frontend components.
    - Configuration of Wildfly, HAproxy, Postgres.
    - Installation of Flyway, Sonatype Nexus, Runners for continuous integration.
    - Installation and configuration of ELK for log management.
    - Installation and configuration of Sonarqube for static code analysis.
2. On-premise maintenance of development, quality, and pre-production environments.
3. Support for deployment and production releases.

Used Technologies: Debian, Flyway, HAProxy, Nexus, Apache, GitLab, GitLab CI, WildFly, SonarQube, PostgreSQL, CentOS, Docker

BACKEND DEVELOPER – CONTINENTAL SAC – 31/10/2020 – 30/11/2021 – PERU

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1. Maintained the Utilex e-commerce platform, developing Python scripts for automated stock updates, resolving production issues, and fixing frontend bugs in Vue.js
2. Led the payment gateway migration to Mercado Pago, ensuring a seamless transition of the checkout and payment flows
3. Integrated Google Analytics e-commerce tracking into the Vue.js frontend to measure user behavior and sales performance
4. Developed Python backend services for sales, marketing campaigns, and frequent customer reporting
5. Implemented a product recommendation section powered by Amazon OpenSearch, improving product discovery for customers
6. Managed the platform's AWS cloud architecture, including serverless and containerized services

Used Technologies: Python, Vue.js, AWS Lambda, AWS Fargate, Amazon RDS, Amazon DynamoDB, Amazon S3, Amazon Cognito, Amazon CloudWatch, Amazon OpenSearch

BACKEND ANALYST – PLATANITOS BOUTIQUE – 30/11/2018 – 30/06/2019 – PERU

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1. Maintained the PHP e-commerce platform, troubleshooting and resolving production issues in web services and databases to ensure service availability
2. Developed Python serverless web services on AWS Lambda as part of the e-commerce migration from a PHP monolith to a serverless architecture
3. Built a Proof of Concept (POC) for in-store visit analytics using Amazon Rekognition (computer vision) to count visitors and estimate demographic data, providing insights into customer traffic

Technologies: Python, PHP, AWS Lambda, Amazon RDS, Amazon DynamoDB, Amazon S3, Amazon Cognito, Amazon CloudWatch, Amazon Rekognition

PROGRAMMER ANALYST – MINISTERIO PUBLICO - FISCALIA DE LA NACION – 31/12/2017 – 30/11/2018 – PERU

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1. Analyzed and developed the backend in Java and the frontend in Angular for the audio and video component of the Electronic File Folder pilot project, enabling multimedia evidence to be attached to digital case files
2. Developed RESTful web services in Java for the Notification Generator, Notification Center, and Electronic Mailbox systems, supporting the digital delivery of official notifications
3. Implemented Docker and Portainer to create and monitor development, QA, and production environments
4. Built CI/CD pipelines with Jenkins, GitLab, and Nexus to automate deployments to QA and production
5. Installed and configured pgBarman to automate PostgreSQL database backups, improving data recovery and reliability

Technologies: Java 8, JPA, Angular 6, JavaScript, RESTful web services, PostgreSQL, WildFly, Apache, HAProxy, Squid Proxy, Docker, Portainer, Jenkins, GitLab, Nexus, pgBarman

DEVELOPER – SUNAT – 01/2017 – 02/2018

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1. Resolution of incidents arising from databases in the Tax Systems Development Division.
2. Resolution of incidents arising from the funds release module in the Tax Systems Development Division.
3. Resolution of incidents arising from the funds retention-perception module in the Tax Systems Development Division.
4. Resolution of incidents arising from the payment cart module in the Tax Systems Development Division.
5. Maintenance and development of functionalities in the funds release module of the Tax Systems Development Division.
6. Maintenance and development of 4GL batch processes in the Tax Systems Development Division.

Used Technologies: Java, Oracle Weblogic, InformixSQL, JSP, Ajax, JavaScript, MyBatis.

PROGRAMMER ANALYST – IRFSAC – 30/09/2015 – 30/11/2016 – PERU

1. Analysis and backend-frontend development of the PhoneStudio product: adaptation and development of functionalities for a radio station using Asterisk software.
2. Analysis and backend-frontend development of an operator interface for a medium-wave transmitter: operation, protection, and display of operating parameters on an embedded PC.
3. Implementation of audio streaming transmitter and receiver based on OpenOB on an embedded PC.
4. Implementation, management, and adaptation of a website using the Joomla tool (www.irfsac.com).

Used Technologies: Joomla, Asterisk, PHP, Raspberry Pi, Python

EDUCATION AND TRAINING

08/03/2011 – 18/12/2015 Lima, Peru
SOFTWARE ENGINEERING UNMSM

02/09/2019 – 30/06/2020 Spain
MASTER ON INTERNET OF THINGS Universidad Complutense de Madrid

28/05/2024 – 03/12/2024
DIPLOMA IN ARTIFICIAL INTELLIGENCE Pontificia Universidad Catolica de Chile

09/09/2025 – CURRENT Barcelona, Spain
MASTER IN ARTIFICIAL INTELLIGENCE Universitat Politècnica de Catalunya

LANGUAGE SKILLS

Mother tongue(s): **SPANISH**

Other language(s):

	UNDERSTANDING		SPEAKING		WRITING
	Listening	Reading	Spoken production	Spoken interaction	
INGLES	C1	C1	C1	C1	C1
PORTUGUESE	B2	B2	B2	B2	B2

SKILLS

Languages and frameworks

Web framework: Django | python | javascript | angular | vuejs | TypeScript | Nodered | php | c++/c | React | NodeJs | java

Databases

Neo4j | redis | PostgreSQL | Dgraph | MySQL | MongoDB | DynamoDB

Operative Systems

Debian | Ubuntu | Windows 10 | Centos | ContikiOS

Other tools

docker | asterisk | elasticsearch | nodered | Nexus | Flyway | Sonarqube | thingspeak | jenkins | gitlabCI

PROJECTS

03/2022 – 10/2022
Monitoring system for Peña fish farm

Implementation of an IOT solution for monitoring physical-chemical parameters in the Peña fish farm, a trout farm:
- Development of proposal for network deployment
- Development of software for ESP32 microcontroller
- Joint work with electronic and chemistry engineers